

Tag and Bag Apparel SOP

One episode tags, folds, bags, seals, and sorts two apparel items. Each garment comes off its stage into the working area, takes one swift tag at its seam from the mock tag gun, is folded into a packet on the table, is slid into a poly bag, is sealed at the flap, and goes onto the spot matching the size on its label. The **first garment** is staged wherever the episode's config puts it, the **second garment** at the right edge, and the first one is worked first.

The order never changes within a garment: stage, tag, fold, bag, seal, tray. No garment is folded before it is tagged, no garment goes into a bag before it is folded, no bag is carried to the tray before its flap is sealed, and no bag goes down until the size on that garment's label has been read.

The left gripper brings the first garment in when it starts on the left or at the front-center, takes the poly bags, and holds whatever the right gripper is working on. The right gripper brings the first garment in when it starts at the front-right, brings the second garment in, and does the tagging, the folding, the insert, the seal, and the placing on the tray. Neither gripper takes the other's stage, the two arms never cross each other, nothing is folded or opened in the air, nothing is turned in the air, and a gripper opens only over the spot the thing in it is going to.

The table is set up in one of three ways. Only the first garment moves; the second garment stage, the bag stack, the tag gun spot, the working area, and the tray are in the same place in all three.

- **Config L:** the first garment is at the left edge.
- **Config M:** the first garment is at the front-center, in front of the working area.
- **Config R:** the first garment is at the front-right corner.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the first garment starts at the left edge (**Config L**), the front-center (**Config M**), or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the first garment's side brings it in — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table for it.
- **Fixed roles:** everything else is the same in all three configs — the right gripper brings the second garment in from the right edge, the left gripper takes the bags and holds, and the right gripper tags, folds, inserts, seals, and places.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the first garment in the start zone for this episode's config, the second garment stage at the right edge, the bag stack at the front left, the tag gun spot right of the working area, the working area at the center, and the tray at the back center.
3. The working area is visible from above, so every fold, every insert, and every seal can be seen.
4. Both arms are at home with grippers open.
5. The table is bare apart from the garments, the poly bags, the mock tag gun, and the tray.
6. The left arm reaches the bag stack, the working area, and (Config L and M) the first garment stage. The right arm reaches the second garment stage, the tag gun spot, the working area, the tray, and (Config R) the first garment stage. Neither arm needs the other's zones.

Materials checklist

1. Two **garments** of soft folded apparel, each lying flat and square: the **second garment** at the second garment stage on the right edge, and the **first garment** in the start zone for this episode's config, the other two zones bare:
 - **Config L:** left edge
 - **Config M:** front-center, in front of the working area, clear of the bag stack
 - **Config R:** front-right corner, in front of the second garment stage and the tag gun spot
2. Every garment lies hem edge toward the front edge, collar edge away, with its **size label** face up and readable from above.
3. The two sizes vary between episodes. Both garments carrying the same size in one episode is normal.
4. Two **poly bags** lie flat in one stack at the front left, both facing the same way, each with its **bag flap** lying back flat and its adhesive strip exposed, clean, and facing up, so the strip meets the bag when the flap is folded over.
5. One **mock tag gun** lies flat on the tag gun spot with its tip pointing toward the working area, loaded with two **swift tags**, handle clear so a gripper can take it from above.

6. One **tray** sits at the back center, empty and square to the table. Its surface carries three **size spots** in a row across it, left to right, with nothing marking them. The spots take S, M, and L in that order, and the order never changes between episodes. Each spot has room for two sealed bags lying flat, one on top of the other.
7. The working area at the center of the table is clear and dry.

Workspace layout

- **First garment stage:** the first garment (input) — left edge (**Config L**), front-center in front of the working area (**Config M**), or front-right corner (**Config R**). One per episode. The zone is bare once the first garment is in the working area.
- **Second garment stage:** right edge, one garment lying flat, the same in all three configs.
- **Bag stack:** front left, two poly bags flat in one stack.
- **Tag gun spot:** the clear patch right of the working area, the mock tag gun lying flat with its tip toward the working area.
- **Working area:** the center of the table, where every tag press, fold, insert, and seal happens.
- **Tray:** back center, three unmarked size spots running left to right across it, taking S, M, and L in that order.

The second garment, the tag gun, and the tray are staged on the right or in the center and the bags on the left; the first garment is brought in by the gripper on its side. No carried thing crosses the midline, and the two arms never cross each other. The left gripper stays left of the right gripper for the whole episode, and an arm with nothing to do waits on its own side, clear of the working area.

Vocabulary

- **Arm assignment:** the left gripper brings the first garment in (Config L and M), takes the poly bags, and holds whatever the right gripper is working on. The right gripper brings the first garment in (Config R), brings the second garment in, works the tag gun, makes both folds, slides the packet into the bag, seals the flap, and places the sealed bag. Neither gripper takes the other's stage and neither takes over the other's work.
- **Garment:** one apparel item. Its **hem edge** is the edge nearest the front edge as it lies in the working area and its **collar edge** is the edge farthest from it. The **first garment** starts in the start zone and is worked first; the **second garment** starts at the right edge.
- **Start zone:** where the first garment lies at the start of the episode — left edge (**Config L**), front-center (**Config M**), or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Size label:** the printed size on the outside of a garment. It is read once, in the working area before the folds, and it decides the size spot.

- **Size spot:** the place on the tray a sealed bag belongs. Nothing marks the spots, so the spot comes from its position across the tray: the left third takes S, the middle third takes M, and the right third takes L. The order is the same in every episode and is never read off the tray.
- **Seam:** a line where two pieces of cloth are joined together by stitches. On a garment it shows as a raised stitched ridge running along an edge.
- **Tag seam:** the side seam on the right edge of the garment as it lies in the working area, level with the middle of that edge. Every tag press is made there and nowhere else.
- **Mock tag gun:** the plastic tool that carries the swift tags. It is held by the handle, taken off a flat surface, carried level with the tip already pointing forward, and laid back down flat with the tip toward the working area. It is never turned in the air.
- **Tag press:** the tip set on the tag seam, pressed against it, held for 2 seconds, and lifted straight off. One press attaches one swift tag and the tag is treated as attached from that press.
- **Poly bag:** one flat bag holding one folded garment. Its **bag mouth** is the open end, its **top wall** is the upper face of the bag above the mouth, and its **bag flap** is the lip beyond the mouth, carrying an exposed adhesive strip on its upper face. The strip is exposed from staging, so no flap is ever peeled during an episode.
- **Sealed bag:** the flap folded back over the bag mouth onto the top wall and pressed along its length, so the adhesive strip has taken hold across the full width, no corner is lifting, and the flap stays down when the bag body is nudged.
- **Folded packet:** the garment after both folds, lying flat on the table with no layer standing up or hanging loose, and no wider than the bag mouth.
- **Crossing:** the two arms overlapping in the air. One gripper passing over, under, or around the other, the two arms touching or stacked, or a gripper reaching past the other arm to get to something. The arms never cross. On a shared thing the left gripper keeps to the left of the right gripper, and when one gripper has to work the other's half, the other lifts clear and withdraws to its own side first.
- **Flat hold:** the left gripper closed and resting on a garment, a packet, or a bag body to keep it from sliding while the right gripper works. During a fold it lifts clear as the incoming layer reaches it, so the fold lands flat.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Lower on:** carrying level over the size spot, lowering straight down until the bag is resting on the tray, and releasing once it is resting. Nothing is dropped from height.
- **Settled:** the thing stays put for 2 seconds after the gripper lifts clear, with nothing sliding, springing open, or hanging over an edge.
- **Release point:** over the working area, inside the bag mouth, on the tag gun spot, or over the size spot the bag is going onto. A gripper opens nowhere else.

Steps

Steps 1 to 6 finish one garment. Run them twice, taking the first garment first, then the second garment. Only the first pick in Step 1 depends on the config: the **left gripper** brings the first garment in from the left edge (Config L) or the front-center (Config M), the **right gripper** from the front-right (Config R). Every other line is the same in all three configs. Step 7 ends the episode once the second sealed bag is on the tray.

Step 1: Stage one garment in the working area

Goal: one garment lying flat and square in the working area with its size label face up.

Look where the first garment is before reaching for it.

- **IF the first garment is at the left edge (Config L):** the **left gripper** closes on the garment at the **first garment stage**, lifts it straight up, carries it level to the **working area**, lays it flat with the hem edge toward the front edge, and releases.
- **IF the first garment is at the front-center (Config M):** the **left gripper** closes on the garment at the **first garment stage**, lifts it straight up, carries it level straight back from the front edge into the **working area**, lays it flat with the hem edge toward the front edge, and releases.
- **IF the first garment is at the front-right (Config R):** the **right gripper** closes on the garment at the **first garment stage**, lifts it straight up, carries it level to the **working area**, lays it flat with the hem edge toward the front edge, and releases.

Then, in all three:

- For the second garment, the **right gripper** does the same from the **second garment stage** at the right edge.
- Read the size label before moving on. It decides the size spot in Step 6.

Take one garment per trip and finish it before the next one comes in. Neither gripper reaches across the table for a garment. The gripper that is not carrying waits on its own side, clear of the working area, so the carrying arm never passes over it.

Check: the garment lies flat and square in the working area, hem edge toward the front edge, no edge folded under itself, and the size label face up and readable. If it lands bunched or folded over itself, lift it clear and lay it down again.

Step 2: Tag the seam

Goal: one swift tag pressed onto the tag seam and the gun back on its spot.

2.1 Take the gun

- The **right gripper** closes on the handle of the **mock tag gun**, lifts it straight up off the tag gun spot, and carries it level to the working area with the tip already pointing forward.

2.2 Press the seam

- The **left gripper** takes a **flat hold** on the garment at its left edge and keeps it for the whole press.
- The **right gripper** sets the tip on the **tag seam**, presses it against the seam, holds for 2 seconds, and lifts straight off.
- Do not drag the tip along the seam, and press nowhere but the tag seam. The left gripper stays on the left edge and the right gripper on the right edge, so the arms never cross over the garment.

2.3 Return the gun

- The **right gripper** carries the gun level back to the **tag gun spot**, lays it down flat with the tip toward the working area, and releases.
- The **left gripper** releases the flat hold once the gun is clear of the garment.

Check: one press was made on the tag seam, the garment is still flat and square with the size label face up, and the gun lies flat on its spot with the tip toward the working area.

Step 3: Fold the garment

Goal: a folded packet lying flat in the working area.

Both folds are made on the table. Never lift the garment to fold it in the air.

3.1 Fold right edge to left edge

- The **left gripper** takes a **flat hold** at the garment's left edge.
- The **left gripper** lifts clear as the incoming edge reaches it and withdraws to its own side, out of the working area, before the right gripper arrives.
- The **right gripper** pinches the middle of the right edge, drags it horizontally left across the garment, and lays it on the left edge, then releases. It reaches the left edge only once the left gripper has withdrawn, and never passes over the left gripper.

Expected state: a narrow rectangle with the folded spine on the left.

3.2 Fold collar edge to hem edge

- The **left gripper** takes a **flat hold** on the left of the near half.
- The **right gripper** pinches the middle of the collar edge, drags it toward the front edge over the near half, and lays it on the hem edge, then releases. It works the right of the packet and does not pass over the left gripper.
- The **left gripper** lifts clear as the incoming edge reaches it and withdraws to its own side.

Check: the packet lies flat with no layer standing up or hanging loose and is no wider than the bag mouth. If a layer stands proud or the packet is too wide for the mouth, open the last fold, lay the layer flat, and fold it again.

Step 4: Bag the garment

Goal: the whole folded packet inside one poly bag with the flap still free.

4.1 Lay a bag out

- The **left gripper** closes on the near edge of the top **poly bag** on the bag stack, lifts it straight up, carries it level to the working area, and lays it flat just beyond the folded packet, away from the front edge, with the **bag mouth** and **bag flap** toward the front edge, then releases.

Take one bag per trip. The **right gripper** waits on its own side while the bag comes in. The flap stays lying back flat with its adhesive strip facing up until Step 5. Nothing is set down on the strip.

4.2 Slide the packet in

- The **left gripper** pinches the **top wall** at the left of the bag mouth and lifts it just enough for the mouth to stand open.
- The **right gripper** closes on the near edge of the **folded packet**, lifts it, brings it level to the open mouth from the right of the left gripper, and slides it in away from the front edge until the whole packet is inside and clear of the mouth, then releases inside the bag and withdraws. It never passes over or under the left gripper.
- The **left gripper** lays the top wall back down flat and releases.

Check: the whole packet is inside the bag with no part of it standing proud of the mouth, the bag lies flat, and the flap is still lying back flat toward the front edge with its adhesive strip facing up and clear. If the packet is proud of the mouth, hold the mouth open again and slide it the rest of the way in.

Step 5: Seal the bag flap

Goal: a sealed bag lying flat in the working area.

- The **left gripper** takes a **flat hold** on the bag body beyond the mouth, on the left of the bag, and keeps it for the whole seal.
- The **right gripper** closes on the near edge of the **bag flap**, folds it away from the front edge over the **bag mouth**, and lays it down on the **top wall** so the adhesive strip meets the bag.
- Press along the flap from one end to the other until the strip has taken hold across the full width and no corner is lifting. The right gripper stays on the flap and does not pass over the left gripper.
- Release the right gripper, then the left.

The flap is folded once and laid down once. Do not lift a stuck flap to place it again, because the strip loses its hold on a second try.

Check: the flap lies flat on the top wall over the bag mouth, the adhesive strip has taken hold across the full width, no corner is lifting, and the flap stays down when the bag body is nudged. If a corner lifts, press along it again.

Step 6: Place the bagged garment on the tray by size

Goal: the sealed bag lying flat on the **size spot** for the size read from that garment, flap face up.

- Take the **size spot** from the size read in Step 1: left third for S, middle third for M, right third for L. Nothing marks the spots, so the spot is counted off from the left end of the tray every time.
- The **right gripper** closes on the near edge of the **sealed bag**, lifts it straight up, carries it level to the **tray**, and lowers it onto that spot until it is resting on the tray or on the bag already there.
- Release once it is resting.

The **left gripper** waits on its own side while the bag is carried, so the arms do not cross on the way to the tray. The bag goes down the way it was sealed, flap face up. Nothing is turned in the air, and no bag is pressed down after it is resting.

Check: the bag lies flat on the size spot for its size, counting from the left end of the tray, flap face up, nothing hangs over the edge of the tray, and the bag already on that spot has not been disturbed.

Step 7: Confirm the two bagged garments and end the episode

- Confirm the tray holds two sealed bags, each lying flat on the size spot for its size with the flap face up, and confirm both garment stages and the bag stack are empty, the mock tag gun lies flat on its spot with the tip toward the working area, and the working area is bare.
- Return both arms home, each to its own side, clear of the tray and the working area and not crossing on the way, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the two sealed bags off the tray and set the tray back at the back center, bare, square to the table, and in the same left to right orientation every time. Nothing marks the spots, so a tray turned end for end silently reverses S and L.
2. Peel each flap back off the bag, lay it flat again with its adhesive strip facing up, and slide the garment out.
3. Shake each garment out flat and check it for tears and for a readable size label. Replace any garment that is torn or whose label cannot be read.

4. Lay the second garment flat at the second garment stage on the right edge, and the first garment flat in the start zone for the next episode's config — left edge (Config L), front-center (Config M), or front-right (Config R) — both hem edge toward the front edge with the size label face up, leaving the other two zones bare.
5. Vary the sizes and which garment goes to which stage between episodes rather than running the same layout every time. The count stays at two garments and two bags.
6. Stack two poly bags flat at the front left, both facing the same way, each flap lying back flat with its adhesive strip facing up and clean. Test each strip with a fingertip and swap in a fresh bag if it has lost its tack, picked up lint, or torn at the fold.
7. Reload the mock tag gun with two swift tags and lay it flat on the tag gun spot with the tip toward the working area.
8. Wipe the working area and confirm the surface is clean and dry. A dusty table lets a garment slide under a flat hold.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config L (the first garment starts at the left edge). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the first garment is not in the start zone for the config; a gripper reaches across the table for the first garment; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the left gripper brings the first garment in in Config L and M, the right gripper in Config R; no arm reaches across the table

for it; the IF line followed is the one for the config on the table).

- **Coaching note:** look where the first garment is before the first reach, then follow that config's IF line through Step 1.

Violation: Wrong order

- **Visible cue:** a garment is folded before the tag press, a packet goes into a bag before both folds are made, a bag is carried to the tray before the flap is sealed, or the tag gun comes out again after the folds have started.
- **SOP rule broken:** Steps 1 to 6, the order within a garment never changes: stage, tag, fold, bag, seal, tray.
- **Coaching note:** stage, tag, fold, bag, seal, tray. Finish the step you are in before you start the next one.

Violation: Garments taken out of order

- **Visible cue:** the right stage garment is worked before the left stage garment is sealed and on the tray, or both garments are brought into the working area before either is finished.
- **SOP rule broken:** Step 1, take the left stage garment first, then the right stage garment, and take one garment per trip.
- **Coaching note:** left stage empties first. One garment in the working area at a time.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the right stage garment, works the tag gun, makes a fold, slides the packet in, seals a flap, or reaches the tray; or the right gripper takes the left stage garment or a poly bag.
- **SOP rule broken:** Steps 1 to 6, the left gripper takes the left stage garment and the bags and holds what the right gripper works on, and the right gripper tags, folds, inserts, seals, and places.
- **Coaching note:** each arm stays on its own side and keeps its own job. Nothing crosses the midline. Reaching past the other arm is scored as **Arms crossed**.

Violation: Arms crossed

- **Visible cue:** one gripper passes over, under, or around the other, the two arms touch or stack in the air, a gripper reaches past the other arm to get to something, or the left gripper ends up right of the right gripper. Most often the right gripper reaches the left edge on the first fold while the left gripper is still on it, or an idle arm is left sitting in the working area while the other arm carries over it.
- **SOP rule broken:** Steps 1 to 7, the two arms never cross, the left gripper stays left of the right gripper, and an arm with nothing to do waits on its own side clear of the working area.
- **Coaching note:** one arm at a time in shared space. If the other arm is in the way, wait for it to withdraw to its own side before you move in.

Violation: More than one garment or bag handled at once

- **Visible cue:** both garments are carried in together, two poly bags come off the stack together, or a garment is moved while the tag gun is still in the other gripper.
- **SOP rule broken:** Steps 1 and 4, one garment per trip and one bag per trip.
- **Coaching note:** one garment, one tag, one bag, one spot. Never bundle a cycle.

Violation: Garment not laid flat and square

- **Visible cue:** the garment is released in the working area bunched, folded over itself, turned across the table instead of hem edge toward the front edge, or with the size label face down, and the tag press is made on it anyway.
- **SOP rule broken:** Step 1, the garment lies flat and square with the hem edge toward the front edge and the size label face up, and a bad landing is corrected by lifting it clear and laying it down again.
- **Coaching note:** square it up before you reach for the gun. Everything after this lands wrong otherwise.

Violation: Tag pressed off the seam

- **Visible cue:** the gun tip meets the middle of the garment, the collar edge, the hem edge, or the left edge instead of the tag seam on the right edge, or more than one press is made on one garment.
- **SOP rule broken:** Step 2.2, one press on the tag seam and nowhere else.
- **Coaching note:** find the seam on the right edge first, then bring the tip down on it.

Violation: Press not held, or the tip dragged

- **Visible cue:** the tip touches the seam and comes away at once, the tip is drawn along the seam while pressed, or the gun is lifted away sideways instead of straight off.
- **SOP rule broken:** Step 2.2, press the tip against the seam, hold for 2 seconds, and lift straight off without dragging.
- **Coaching note:** down, hold two, straight up. The tag does not seat on a tap.

Violation: Garment not steadied during the tag press

- **Visible cue:** the left gripper is off the garment while the gun presses, or the garment slides, rucks, or lifts with the gun on the way out.
- **SOP rule broken:** Step 2.2, the left gripper takes a flat hold at the left edge and keeps it for the whole press.
- **Coaching note:** the flat hold goes on before the tip touches and comes off after the gun is clear.

Violation: Tag gun left off its spot or laid down turned

- **Visible cue:** the gun is parked on the working area, on a garment, or on a stage between garments, is still in the gripper while a fold or an insert is made, or is laid back on its spot with the tip pointing away from the working area.
- **SOP rule broken:** Step 2.3, carry the gun back to the tag gun spot and lay it down flat with the tip toward the working area before anything else is done.
- **Coaching note:** the gun goes home after every press, tip forward, so the next pick is clean.

Violation: Garment lifted off the table to fold it

- **Visible cue:** the garment leaves the table during a fold, an edge is carried over in the air, or the packet is shaped while hanging from a gripper.
- **SOP rule broken:** Step 3, both folds are made on the table and the garment is never lifted to fold it.
- **Coaching note:** drag the edge across the table. Nothing folds in the air.

Violation: Folds made in the wrong order or direction

- **Visible cue:** the collar to hem fold is made before the right to left fold, an edge is dragged the opposite way, or only one fold is made before the bag comes out.
- **SOP rule broken:** Step 3, fold the right edge left onto the left edge first, then the collar edge toward the front edge onto the hem edge.
- **Coaching note:** right to left, then far to near. Two folds, always in that order.

Violation: Flat hold missed or left under the fold

- **Visible cue:** the left gripper never lands on the garment during a fold and the bottom layer drags along with the incoming edge, or the gripper is still on the garment when the incoming layer lands on top of it.
- **SOP rule broken:** Step 3, the left gripper takes a flat hold and lifts clear as the incoming layer reaches it.
- **Coaching note:** hold it down through the drag, then get out of the way just before it lands.

Violation: Bag laid out the wrong way round

- **Visible cue:** the bag is laid with the mouth and flap pointing away from the front edge or off to a side, or it is laid between the front edge and the packet instead of just beyond it.
- **SOP rule broken:** Step 4.1, lay the bag flat just beyond the folded packet, away from the front edge, with the bag mouth and flap toward the front edge.
- **Coaching note:** mouth toward you, bag beyond the packet. The packet only ever travels away from you.

Violation: Bag mouth not held open during the insert

- **Visible cue:** the packet is pushed at a closed bag, the left gripper is off the top wall while the packet goes in, or the mouth collapses and the packet rides over the bag instead of into it.
- **SOP rule broken:** Step 4.2, the left gripper lifts the top wall at the mouth so the mouth stands open, and holds it while the packet slides in.
- **Coaching note:** open it first, then load it. A flat bag will not take a packet.

Violation: Packet dropped in or left proud of the mouth

- **Visible cue:** the gripper opens above the bag and the packet falls in, or the gripper releases with part of the packet still standing out of the mouth and the flap is sealed over it anyway.
- **SOP rule broken:** Step 4.2, slide the packet in away from the front edge until the whole packet is inside and clear of the mouth, then release inside the bag.
- **Coaching note:** slide it all the way in and check the mouth before you let go.

Violation: Flap not sealed

- **Visible cue:** the flap is left lying back, folded over but never pressed, pressed only at one end, or lifting at a corner when the bag is carried, and the bag goes to the tray anyway.
- **SOP rule broken:** Step 5, fold the flap over the bag mouth onto the top wall and press along it from one end to the other until the adhesive strip has taken hold across the full width with no corner lifting.
- **Coaching note:** press the whole length, then nudge the bag to see if it holds.

Violation: Stuck flap lifted and laid down again

- **Visible cue:** the flap is peeled back off the top wall after it has taken hold and folded down a second time, or it is worked loose and repositioned before the press.
- **SOP rule broken:** Step 5, the flap is folded once and laid down once, and a stuck flap is not lifted to place it again.
- **Coaching note:** line the fold up before it lands. The strip will not hold a second time.

Violation: Bag not held during the seal

- **Visible cue:** the left gripper is off the bag body while the flap is folded or pressed, or the bag slides, rucks, or lifts under the press.
- **SOP rule broken:** Step 5, the left gripper takes a flat hold on the bag body and keeps it for the whole seal.
- **Coaching note:** anchor the body first, release it last.

Violation: Bag put on the wrong spot

- **Visible cue:** a sealed bag goes onto a spot whose position does not match the size on that garment, counting S, M, and L from the left end of the tray, or the bag goes down without the

size having been read while the garment lay flat.

- **SOP rule broken:** Steps 1 and 6, read the size label in the working area and lower the bag onto the size spot for that size, left for S, middle for M, right for L.
- **Coaching note:** read the size while the garment is still flat, and say it to yourself before you reach for the tray.

Violation: Bag dropped onto the tray, pressed down, or laid flap face down

- **Visible cue:** the gripper opens above the tray and the bag falls, the bag is pressed down after it is resting, it lands flap face down, or it hangs over the edge of the tray.
- **SOP rule broken:** Step 6, lower the bag until it is resting, release once it is resting, flap face up, and never press a bag down after it is resting.
- **Coaching note:** take it all the way down, then let go. Leave it how it lands.

Violation: Turned in the air

- **Visible cue:** a garment, a packet, a bag, or the tag gun is rotated or flipped between the lift and the set down, so it lands facing a different way than it was carried.
- **SOP rule broken:** Steps 1 to 6, nothing is turned in the air; the orientation set on the table is the orientation it lands in.
- **Coaching note:** set it down and square it on the table. The wrist does not have the range to turn it in transit.

Violation: Dragged instead of lifted

- **Visible cue:** a garment, a bag, or the tag gun slides along the table still touching it, with no daylight under it in transit. The fold drags in Step 3 are not this.
- **SOP rule broken:** Steps 1, 2, 4, and 6, lift straight up until there is daylight before moving sideways.
- **Coaching note:** the lift is its own motion, not the start of the carry.

Violation: Regripped in the air

- **Visible cue:** the gripper shifts, rolls, or seats its hold again on a garment, a packet, a bag, or the tag gun without setting it down first.
- **SOP rule broken:** Steps 1 to 6, set it down before taking a new hold.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over a garment stage, over the bag stack, over the tray while carrying a garment or a loose bag, or over open table.
- **SOP rule broken:** Steps 1 to 6, a gripper opens over the working area, inside the bag mouth, on the tag gun spot, or over the size spot the bag is going onto, and nowhere else.

- **Coaching note:** know where it lands before you open the gripper.

Violation: Placement corrected by shoving

- **Visible cue:** a gripper nudges, pats, or slides a garment, a packet, a bag, or the tag gun into position after releasing it, instead of lifting it clear and placing it again.
- **SOP rule broken:** Steps 1 to 6, correct a bad placement by lifting the thing clear and placing it again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a garment, a bag, or the tag gun falls to the table or the floor short of its spot, the tray is struck and shifts, a sealed bag already on the tray is knocked off, the bag stack is scattered, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 6, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a garment still on a stage, a bag unsealed or still in the working area, a loose bag on the stack, the tag gun off its spot, an arm away from home, an arm still over the tray or the working area.
- **SOP rule broken:** Step 7, confirm the two sealed bags on their matching spots and the empty stages, stack, and working area; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective garment:** torn mid episode, or a size label that cannot be read once it is flat in the working area. Replace it before the next episode.
- **Tag gun fault:** the gun runs out of swift tags before the second garment, the tip breaks, or the handle is too slick for the gripper to lift off a flat surface. Reload or replace it before the next episode.
- **Defective poly bag:** a torn bag, a mouth that will not stand open, a flap torn at the fold, or an adhesive strip that will not take hold under a correct press because it has gone dead or

picked up lint. Swap in a fresh bag before the next episode.

- **Tray shifts out of its zone** during the episode, is turned end for end so the spot order runs the wrong way, or is too small to take both sealed bags without one overhanging an edge. Square the tray up before the next episode.
- **Garment or bag slides** under a correct flat hold because the table is dusty or slick. Wipe the table before the next episode.

Annotation subtasks (from SOP)

1. Move one garment from a stage to the working area
2. Take the mock tag gun off its spot
3. Press one swift tag onto the tag seam
4. Return the mock tag gun to its spot
5. Fold one garment right edge to left edge
6. Fold one garment collar edge to hem edge
7. Lay one poly bag out in the working area
8. Slide one folded packet into the poly bag
9. Seal the bag flap
10. Place one sealed bag by size on the tray
11. Return both arms home and end the episode